

Tony Aizize

Singapore • aibiertony@gmail.com • +65 ***-8060 • linkedin.com/in/tony007 github.com/Aibier •
globalpeacelove.com/about-me

Results-driven **Software Engineering Manager & de facto Software Architect** with 10+ years of experience building scalable payment infrastructure, crypto integrations, and fintech innovation. Led 50+ bank, PSP, and stablecoin integrations, delivering resilient money-movement systems that power cross-border transactions across APAC, EMEA, and the US.

With proven expertise in **event-driven microservices, secure wallet integrations, and cloud-native scaling**, I bridge deep technical architecture with business strategy — enabling high-performance platforms for the next wave of financial innovation. Passionate about mentoring engineering teams, shaping technical culture, and driving adoption of blockchain and stablecoin-powered solutions.

TECHNICAL EXPERTISE

- **Languages:** Golang, PHP, Python, Perl, Java (Spring Boot, Spring), JavaScript/TypeScript
- **Architecture:** Microservices, API Design, Event-Driven Systems, Distributed Ledgers, AWS Cloud
- **Tech Stack:** Docker, CI/CD, PostgreSQL, MySQL, MongoDB, Redis, Kafka, AWS SQS
- **Certifications:** AWS Certified Solutions Architect (Associate), Certified Big Data Professional (NUS)
- **Graduate Studies:** Scalable Systems, Securing Ubiquitous Systems (NUS)

LEADERSHIP & STRATEGY

- **Team Leadership:** Managed & scaled teams of 10+ engineers in Core Payments & FX
- **Strategic Execution:** Market expansion via stablecoin and fiat integrations
- **Stakeholder Alignment:** Translating crypto/fintech architectures into business impact
- **Culture:** Driving engineering best practices, performance management, and career growth

Mission-driven to build the next generation of crypto-native payment platforms, scale decentralized finance solutions, and deliver engineering excellence at global scale.

EDUCATION

Master's Degree in Software Engineering | Jan 2021 – Jan 2023 National University of Singapore
Architecting Software Solutions, Platform Engineering, DevOps Engineering, Cloud Native Solution Design, Big Data Engineering

Graduate Diploma in System Science | Jul 2015 – Nov 2016 Institute of System Science, NUS, Singapore

Bachelor's Degree in Management Science and Engineering | Sep 2011 – Jul 2015 Central University of Finance and Economics, Beijing, China
Mathematics, Computer Science

WORK EXPERIENCE

Software Engineering Manager (Payment & FX Infra)

Mar 2025 – Present

Aspire FT · Singapore

Payment Infrastructure & Technical Leadership

- Lead payment (banks, payment providers) and stablecoin integrations across APAC and globally.

- Led the design and development of scalable payment infrastructure, optimizing money movement flows and ensuring high system efficiency.
- Spearheaded the setup of payment technology infrastructure in Hong Kong, leading debit account payouts, FX transfers, and payout/payin integrations.
- Designed and built a microservices-based Payment Service Architecture, enabling seamless market expansion and preparing tech infrastructure for new regions.

Team Leadership & Engineering Excellence

- Led a team of 10 engineers, fostering technical growth and career development.
- Established engineering best practices, improving system reliability, performance, and security compliance.
- Managed end-to-end payment flows, ensuring compliance with regulatory standards and optimizing transaction monitoring.
- Collaborated with product managers and business stakeholders to align engineering initiatives with strategic goals.

Payment Architect & Engineering Manager

Jul 2022 – Dec 2024

Aspire Financial Technologies · Singapore

Architecture & System Design

- **Next-Gen Payment Service:** Architected a hybrid platform combining Monolith and Microservices to optimize scalability, enabling seamless handling of 13+ currencies (SGD, USD, EUR, GBP) across APAC and the US.
- **Distributed Systems:** Designed a Redis-based rate limiter (sliding window & token bucket algorithms in Golang) to manage 500+ TPS, reducing API throttling incidents by 65%.
- **API & Integrations:** Led development of RESTful APIs for merchant payments and gRPC services for inter-microservice communication, enhancing cross-border transaction reliability.

Technical Implementation

- **API Maintainer for Go Projects:** Reviewed code, drove technical decisions, and built common libraries.
- **Queue Frameworks:** Built a prioritization system using AWS SQS & Kafka, improving transaction throughput by 30% and reducing failed payments by 25%.
- **Infrastructure as Code:** Deployed AWS ECS, RDS, and CI/CD pipelines to automate deployments, reducing release cycles from 2 hours to 15 minutes.
- **Performance Optimization:** Instrumented Datadog dashboards and ELK stack for real-time monitoring, achieving 99.99% system uptime.

Leadership & Team Management

- **Team Leadership:** Directed a 9-member engineering team managing SGD/USD/EUR/GBP FX transfers, reducing currency conversion costs by 18% through dynamic routing.
- **Talent Development:** Hired and mentored 15+ engineers, instituting code review practices with SonarQube that cut critical bugs by 25%.
- **Market Expansion:** Spearheaded technical infrastructure setup in Singapore, Hong Kong, and the USA for debit account issuance, enabling compliance with local regulations and entry into 3 new markets.

Compliance & Scalability

- **Regulatory Alignment:** Engineered a Transaction Monitoring Tool to automate fraud detection, ensuring adherence to MAS and HKMA standards.

- **QA & Code Quality:** Enforced rigorous testing frameworks, reducing production incidents by 40% post-deployment.

Senior Software Engineer Team Lead
Aspire Financial Technologies · Singapore

Nov 2021 – Jul 2022

- Launched Configuration as a Service (CaaS), enabling zero-code market expansions into 5+ countries, cutting deployment time by 70%.
- Led an onboarding development team of 5 engineers.
- Revamped onboarding flows, improving customer activation rates by 35%.

Senior Software Development Engineer (Payments)
Thunes · Singapore

Aug 2020 – Nov 2021

- Migrated legacy Perl systems to Golang, boosting transaction throughput by 60%+.
- Integrated 10+ APAC payment providers (e.g., Alipay, DBS, GrabPay, CZBank) into a unified cross-border platform, supporting 2M+ daily transactions across APAC markets.
- Led integration of GrabPay and TikTok Influencer Pay, reducing integration timelines by 50% through custom API bridges and routing logic.
- Refactored and migrated 30+ APAC integrations from Perl to Golang within 7 months.
- Built a Transaction Tracing API with Elasticsearch-backed log indexing and a human-readable tracing dashboard.

Software Engineer
DocDoc Pte Ltd · Singapore

Jun 2019 – Aug 2020

- Optimized REST APIs for a doctor discovery platform, achieving 99.9% uptime on AWS.
- Implemented CI/CD pipelines, reducing deployment cycles from 2 hours to 15 minutes.

Full Stack Developer
Golden Equator Consulting · Singapore

Dec 2016 – Jun 2019

- Developed REST APIs for Spectrum Mobile App.
- Built Access Door Control System (Brivo Integration).
- Developed React/Redux UI components and integrated Stripe payments for event space platform.
- Built e-commerce websites; managed group server stack: Python, Java, Node.js, React.js, PostgreSQL, MySQL, MongoDB.
- **Applications:** Cerebro Platform APIs (Django REST Framework), Spectrum Global Mobile App (REST API), Golden Equator Group website.

Freelance Web Developer (Part-Time)
Besttours Singapore · Singapore

Sep 2016 – Nov 2016

- Built an e-commerce website with backend APIs and frontend components.
- Built a Kiosk (Ticketing Machine) deployed at Fragrance Hotel Bugis, Singapore.
- Technologies: PHP, CSS3, HTML5, Bootstrap, Angular 2, Electron.

Full Stack Developer (Intern – Full-Time)
Besttours Singapore · Singapore

Apr 2016 – Sep 2016

- Developed and maintained Singapore Changi Airport website frontend.
- Involved in developing a One-Millionaire project; maintained Changi Airport website using CMS (AEM).
- Technologies: PHP Laravel, CSS3, HTML5, Bootstrap.

SKILLS

Category	Details
Databases	PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, CockroachDB (RDBMS & NoSQL)
Architecture	Microservices, Distributed Systems, Event-Driven Architecture, CQRS, Event Sourcing
APIs	RESTful APIs, gRPC, Webhook-driven systems
Languages	Golang, PHP, Perl, Python, TypeScript, Java
Infrastructure	AWS (ECS, RDS, Lambda, SQS, CloudWatch), Docker, Kubernetes, CI/CD
Observability	Datadog, Prometheus, Grafana, ELK Stack
Messaging	Apache Kafka, RabbitMQ, AWS SQS

PROJECTS

Payment Infrastructure

Feb 2022 – Dec 2024

- Designed and built Payout Service, Payin Service, and Webhook Service at Aspire Financial Technologies.
- Led a team of 10 engineers (senior and lead-level), covering architecture, delivery, and hiring.
- Delivered 50+ bank and payment provider integrations across APAC, EMEA, and the US, handling \$50M+ monthly transaction volume.
- Engineered event-driven flows via Kafka for near real-time processing at 500+ TPS with 99.99% system uptime.

LANGUAGES

English (Fluent) • Chinese (Native)